

Day 108 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in General Science (10): Gaganyaan's Final-Phase Testing & Astronaut Training

READING MATERIAL

Covers Gaganyaan's final-phase testing milestones and the astronaut-designates' training progress, extending the Day-78 Gaganyaan coverage with new details.

Q: What test did ISRO and DRDO conduct on 19 February 2026 for the Gaganyaan programme, and what did it evaluate?

A: The final qualification level load test — evaluating high-speed aerodynamic loads and changes in the ballistic trajectory of the capsule.

Q: Which key vehicle/systems are in final stages of testing and integration for Gaganyaan?

A: The Human-rated LVM3 launch vehicle, the Crew Escape System, and the Crew Module and Service Module.

Q: How many Indian Air Force pilots have been selected as astronaut-designates for Gaganyaan, and what is their training status?

A: 4 IAF pilots — they have completed training in RUSSIA and are now undergoing further mission-specific training in India, with continuous assessment of health, psychological fitness, and simulation-based operational readiness.

Q: What recovery trials have been conducted for the uncrewed Gaganyaan mission?

A: Recovery trials with the Indian Navy, with more sea recovery simulations planned — preparing for the capsule's safe recovery after splashdown.

Q: What is Gaganyaan's overall current timeline, per the latest schedule?

A: The uncrewed Gaganyaan-1 (G1) test flight is scheduled for the second half (H2) of 2026; the first CREWED human spaceflight is now scheduled for the first quarter of 2027.

MOCK TEST (WITH ANSWERS)

1. ISRO and DRDO conducted Gaganyaan's final qualification level load test on:

- A. 19 January 2026
- B. 19 February 2026 ✓**
- C. 19 March 2026
- D. 19 April 2026

Explanation: 19 February 2026. [hard, web-research]

2. The 19 February 2026 load test primarily evaluated:

- A. Solar panel efficiency
- B. High-speed aerodynamic loads and ballistic trajectory changes ✓**
- C. Communication signal strength
- D. Fuel consumption rates

Explanation: High-speed aerodynamic loads and ballistic trajectory changes. [medium, web-research]

3. How many IAF pilots have been selected as astronaut-designates for Gaganyaan?

- A. 2
- B. 3
- C. 4 ✓**
- D. 6

Explanation: 4. [medium, web-research]

4. Where did Gaganyaan's astronaut-designates complete their initial training?

- A. United States
- B. Russia ✓**
- C. France
- D. Japan

Explanation: Russia. [medium, web-research]

5. Gaganyaan's first CREWED human spaceflight is now scheduled for:

A. H2 2026

B. Q1 2027 ✓

C. Q4 2027

D. 2028

Explanation: Q1 (first quarter) 2027. [medium, web-research]

Part B – Aptitude & Mental Ability: Puzzles

READING MATERIAL

Puzzles is 1 of 6 named Reasoning skills. The genre overlaps with Logical Reasoning, so this topic focuses on archetypes NOT already covered there: comparison chains, age puzzles, family-relation puzzles, and linear seating arrangements — all extremely common TNPSC puzzle shapes.

Q: Method — comparison chain (not all four compared directly, must chain through clues): D is taller than C but not as tall as B. C is taller than A. Who is the tallest?

A: Chain it: $B > D > C > A$. So B is the tallest. The key technique: write each clue as an inequality, then merge them into one ordered chain rather than trying to compare all pairs mentally.

Q: Method — classic age puzzle: A father is 4 times as old as his son. In 20 years, he will be twice as old as his son. Find their current ages.

A: Let son's current age = x , father = $4x$. In 20 years: father = $4x+20$, son = $x+20$. Condition: $4x+20 = 2(x+20) \rightarrow 4x+20 = 2x+40 \rightarrow 2x=20 \rightarrow x=10$. Son is 10, father is 40.

Q: Method — family-relation puzzle: Pointing to a photograph, a man says, 'She is the daughter of my mother's only son.' How is the woman in the photo related to the man?

A: His daughter. 'My mother's only son' = the man himself (since he's referring to his own mother, and he's an only son). So 'her father' = the man, making the woman his daughter.

Q: Method — linear seating arrangement: Five friends P, Q, R, S, T sit in a row. Q is immediately to the right of P. R is at one end. S is between Q and T. T is not adjacent to P. Find the order.

A: Work from the most fixed clue first: R is at an end — try R at the left end. P-Q must be adjacent with Q right of P, and S is between Q and T, with T not adjacent to P. Testing R,P,Q,S,T: P-Q adjacent ✓, S between Q&T ✓, T not adjacent to P (T is at position 5, P at position 2, not adjacent) ✓, R at an end ✓. Order: R, P, Q, S, T.

Q: Method — distribution/sharing puzzle: A sum of money is divided among A, B, and C so that A gets twice as much as B, and B gets thrice as much as C. If C gets ₹600, find the total sum.

A: $C=600$. $B=3 \times 600=1800$. $A=2 \times 1800=3600$. Total = $3600+1800+600 = 6000$.

MOCK TEST (WITH ANSWERS)

1. P is shorter than Q but taller than R. Q is shorter than S. Who is the tallest?

- A. P
- B. Q
- C. R
- D. S ✓**

Explanation: Chain: $S > Q > P > R$. S is tallest. [medium, authored]

2. A mother is 3 times as old as her daughter. In 15 years, she will be twice as old as her daughter. Find the daughter's current age.

- A. 10
- B. 12
- C. 15 ✓**
- D. 18

Explanation: Daughter= x , mother= $3x$. $3x+15=2(x+15) \rightarrow 3x+15=2x+30 \rightarrow x=15$. [medium, authored]

3. Pointing to a man, a woman says, 'He is the son of my father's only daughter.' If the woman has no siblings, how is the man related to her?

- A. Her son ✓**
- B. Her brother
- C. Her nephew
- D. Her father

Explanation: 'My father's only daughter' = the woman herself (she has no siblings). So the man is her son. [medium, authored]

4. ₹7,000 is divided among A, B, C so that A gets twice as much as B, and B gets twice as much as C. Find C's share.

- A. ₹1,000 ✓**
- B. ₹1,200

C. ₹1,500

D. ₹2,000

Explanation: Let $C=x$, $B=2x$, $A=4x$. $\text{Total}=7x=7000 \rightarrow x=1,000$. [medium, authored]

5. Four friends sit in a row. B is immediately right of A. C is at the left end. D is not adjacent to A. Find the correct order.

A. C, A, B, D ✓

B. A, B, C, D

C. C, D, A, B

D. D, A, B, C

Explanation: C at left end; A-B adjacent with B right of A; D not adjacent to A. Order C,A,B,D satisfies all three conditions. [medium, authored]

Part C – General English: Poem: "Life" — Henry Van Dyke

READING MATERIAL

A reflective poem about living life with courage, purpose, and without regret or fear of the future. Real exam evidence confirms a specific quoted line's authorship; web research fills in the figure-of-speech detail, including two devices (Litotes, Symbolism) not seen elsewhere in this syllabus.

Q: Real exam: "Still seeking what I sought when but a boy, new friendship, high adventure and a crown." Who wrote this?

A: Henry Van Dyke, in "Life" — confirmed via a real TNPSC question testing this exact line's authorship against three other poet options.

Q: What is the poem's central theme?

A: Living life forward-facing, with courage and clear purpose — neither rushing nor dragging, not mourning losses, and not holding back out of fear of an unknown future.

Q: Who was Henry Van Dyke?

A: An American author (1852-1933) — earned degrees from Princeton, served over 20 years as a Presbyterian minister, then returned to Princeton as a professor of English for nearly 20 years.

Q: Identify the figure of speech: "Let me but live my life from year to year."

A: Repetition (the phrase structure repeats a forward-moving, cyclical idea) — cited as Repetition rather than the other options in a real-style question.

Q: Identify the figure of speech: "Not mourning for the things that disappear."

A: Litotes — an understatement that affirms something by negating its opposite (saying 'not mourning' to actually emphasize calm acceptance, rather than stating the acceptance directly).

Q: Identify the figure of speech: "So let the way wind up the hill or down."

A: Personification — 'the way' (the path of life) is treated as something that can actively 'wind', given an active, almost willful quality.

Q: Which phrase in the poem is the clearest example of Symbolism?

A: "The road's last turn" — symbolizing the end of life/journey, rather than a literal physical road.

MOCK TEST (WITH ANSWERS)

1. "Still seeking what I sought when but a boy, new friendship, high adventure and a crown" was written by:

- A. Khalil Gibran
- B. H.W. Longfellow
- C. Henry Van Dyke ✓**
- D. John Masefield

Explanation: Henry Van Dyke — confirmed via a real TNPSC question. [hard, question-bank-2024]

2. "Let me but live my life from year to year" is cited as an example of:

- A. Repetition ✓**
- B. Simile
- C. Metaphor
- D. Oxymoron

Explanation: Repetition. [medium, web-research]

3. "Not mourning for the things that disappear" is an example of:

- A. Litotes ✓**
- B. Apostrophe
- C. Irony
- D. Hyperbole

Explanation: Litotes — understatement via negation. [hard, web-research]

4. "So let the way wind up the hill or down" uses which figure of speech?

- A. Simile
- B. Metaphor

C. Personification ✓

D. Alliteration

Explanation: Personification — the 'way' given the active quality of winding. [medium, web-research]

5. Which phrase is the clearest example of Symbolism in the poem?

A. High adventure

B. Year to year

C. Dim past

D. The road's last turn ✓

Explanation: "The road's last turn" symbolizes life's end. [hard, web-research]

6. Henry Van Dyke, in addition to being a poet, spent over 20 years working as a:

A. Lawyer

B. Presbyterian minister ✓

C. Physician

D. Politician

Explanation: A Presbyterian minister, before returning to Princeton as an English professor. [medium, web-research]